

Gauge-Invariant Magnetic Holonomy in Exact Theta-Graph Scattering

Hénon Route-A Working Series, C138

24 August 2026

Abstract

We equip the exact theta metric graph with a two-dimensional magnetic-flux family. Its directed-bond evolution on $\mathbb{C}^6$ is unitary, invariant under a common phase shift, and obeys an antiunitary identity that sends α to $-\alpha$. A block reduction gives the complete Laurent secular determinant in degrees $\rho^0, \rho^2, \rho^4, \rho^6$. Primitive walks retain signed Kirchhoff amplitudes and oriented winding monomials, although the full determinant is even under flux reversal. Exact zero-, π -, and $\pi/2$ -flux controls separate these statements. This is a source-owned unitary/scattering candidate, not a target spectral or arithmetic match.

1 Frozen magnetic graph

Join vertices L, R by edges of lengths $\ell = (1, 2, 3)$. At each vertex use

$$C = \frac{2}{3}\mathbf{1}\mathbf{1}^T - I_3, \quad S = \begin{pmatrix} 0 & C \\ C & 0 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & I_3 \\ I_3 & 0 \end{pmatrix}.$$

Directed bonds are ordered $(+1, +2, +3, -1, -2, -3)$. For real k and $\alpha = (\alpha_1, \alpha_2, \alpha_3)$, set

$$P_\alpha(k) = \text{diag} \left((e^{i(k\ell_j + \alpha_j)/2})_{j=1}^3, (e^{i(k\ell_j - \alpha_j)/2})_{j=1}^3 \right), \quad U_\alpha(k) = P_\alpha(k) S P_\alpha(k).$$

The symmetric split accumulates one metric length and one signed magnetic phase per traversed bond. Here $\alpha \in \mathbb{R}^3$ is a chosen lift. If α_j is increased by 2π , the two orientations of edge j acquire a common sign E_j , so $U_{\alpha+2\pi e_j} = E_j U_\alpha E_j$. Thus the unitary equivalence class and determinant descend to the flux torus, although the displayed lifted matrix need not be entrywise periodic.

Theorem 1 (unitary gauge and reversal). *$U_\alpha(k)$ is unitary. A common shift $\alpha \mapsto \alpha + c\mathbf{1}$ leaves it unchanged. With coefficient conjugation K and $\Theta = JK$,*

$$\Theta U_\alpha(k) \Theta^{-1} = U_{-\alpha}(k)^{-1}.$$

Proof. $C^2 = I_3$, hence S is real orthogonal; $P_\alpha(k)$ is diagonal unitary. For $D_c = \text{diag}(e^{ic/2}I_3, e^{-ic/2}I_3)$, one has $P_{\alpha+c\mathbf{1}} = D_c P_\alpha$ and $D_c S D_c = S$. Since the diagonal matrices commute, $U_{\alpha+c\mathbf{1}} = P_\alpha D_c S D_c P_\alpha = U_\alpha$.

For reversal, $\overline{J P_\alpha(k) J} = P_\alpha(-k)$ and $JSJ = S$. Thus $\overline{J U_\alpha(k) J} = U_\alpha(-k)$. On the other hand, $P_{-\alpha}(k)^{-1} = P_\alpha(-k)$ and $S^{-1} = S$, so $U_{-\alpha}(k)^{-1} = U_\alpha(-k)$. This proves the identity and also shows why holding α fixed is generally false. $\square$

2 Complete Laurent determinant

Put $x_j = e^{ik\ell_j}$, $q_j = e^{i\alpha_j}$ and

$$X_+ = \text{diag}(x_j q_j), \quad X_- = \text{diag}(x_j / q_j).$$

Sylvester's identity first moves the two half phases together: $\det(I_6 - \rho P_\alpha S P_\alpha) = \det(I_6 - \rho S P_\alpha^2)$. The latter matrix has off-diagonal blocks $-\rho C X_-$ and $-\rho C X_+$, so a block determinant gives

$$D(\rho; x, q) = \det(I_3 - \rho^2 C X_- C X_+) = 1 - \rho^2 T_1 + \rho^4 T_2 - \rho^6 (x_1 x_2 x_3)^2,$$

where, writing $Q_{ij} = q_i/q_j + q_j/q_i$ and letting k denote the complement of i, j ,

$$\begin{aligned} T_1 &= \frac{1}{9} \sum_i x_i^2 + \frac{4}{9} \sum_{i < j} x_i x_j Q_{ij}, \\ T_2 &= \frac{1}{9} \sum_{i < j} x_i^2 x_j^2 + \frac{4}{9} \sum_{i < j} x_i x_j x_k^2 Q_{ij}. \end{aligned}$$

The block reduction proves at once that every odd power of ρ vanishes; the displayed constant, quadratic, quartic, and sextic terms are therefore the complete Laurent polynomial, not a truncation. Only ratios of the q_j occur, agreeing with the operator-level common-phase identity in Theorem 1. Since every Q_{ij} is fixed by $q \mapsto q^{-1}$, the full determinant is also invariant under flux reversal.

At zero flux and $(x_1, x_2, x_3) = (t, t^2, t^3)$, $\rho = 1$, the result is

$$\begin{aligned} &-\frac{1}{9}(t-1)^3(t+1)(t^2+1)(t^2+t+1) \\ &\quad \times (3t^2-2t+3)(3t^2+5t+3), \end{aligned} \tag{1}$$

exactly the C133 specialization.

3 Oriented primitive walks

For a closed directed walk p , define $m_j(p) = N_{+j}(p) - N_{-j}(p)$. Closure gives $\sum_j m_j = 0$, and the ordered weight is

$$A_p \rho^{n_p} e^{ikL_p} \prod_j q_j^{m_j(p)}.$$

Thus the primitive germ is the product of $1 - A_p \rho^{n_p} e^{ikL_p} q^{m(p)}$. The two-step walk $(+i, -j)$ with $i \neq j$ has amplitude $4/9$ and phase q_i/q_j ; its reverse has q_j/q_i . The individual terms distinguish orientation. The complete determinant is inversion-even only after reverse walks are paired: reversal preserves A_p, L_p, n_p and sends the winding vector $m(p)$ to $-m(p)$. Thus determinant evenness and orientation-sensitive ledger data are compatible statements at different aggregation levels.

4 Controls, receipts, and boundary

Flux $q = (-1, 1, 1)$ changes the ρ^2 coefficient by $16x_1(x_2 + x_3)/9$: the two ratios involving q_1 change from 2 to -2 . This proves that common gauge invariance does not make nontrivial flux invisible. At $k = 0$, $\alpha = (\pi/2, 0, 0)$, the correct $\alpha \mapsto -\alpha$ reversal defect vanishes. Holding α fixed instead gives eight nonzero entries and squared Frobenius norm $64/9$. Replacing $2/3$ by $1/2$ destroys vertex unitarity; changing the reverse length of edge three from 3 to 4 destroys bond reversal.

The exact ledger reconstructs 14,760 rooted walks and 1,905 primitive cycles through period eight. An independent checker closes every field, a separate symbolic run performs 197 exact checks, replay is byte-identical, and 44 repaired-hash plus one stale-hash mutations are rejected.

The strict verdict is

(A1_WEAK, A2_FAIL, A3_FAIL, A4_UNITARY_OR_SCATTERING_CANDIDATE).

There is no target divisor, prime or zero table, functional equation, arithmetic Euler factor, root number, automorphy claim, or Hilbert–Pólya operator. Route B is unauthorized. The active firewall is

NO_BAD_EULER_OR_ROOT_NUMBER.

5 Conclusion

C138 turns one exact scattering graph into a gauge-controlled magnetic family whose determinant and oriented paths are both source-owned. External target identification remains unsupported.